

Name: \_\_\_\_\_

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**Problem 1 [10 pts] Which coin?**

A drawer holds two coins. Coin G is fair; coin H is loaded and lands heads with probability 0.7. You pick coin H with probability 0.25, flip it four times, and record the number of heads  $K$ .

| $k$ | $P(K = k \mid \text{fair})$ | $P(K = k \mid \text{loaded})$ |
|-----|-----------------------------|-------------------------------|
| 0   | 0.0625                      | 0.0081                        |
| 1   | 0.2500                      | 0.0756                        |
| 2   | 0.3750                      | 0.2646                        |
| 3   | 0.2500                      | 0.4116                        |
| 4   | 0.0625                      | 0.2401                        |

You see  $K = 3$  heads. What is the probability that you picked the loaded coin?